

PARTIAL FRACTIONS

- **Polynomial**: An expression $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ is called as polynomial in n of degree n if $a_n \neq 0$.
- Remainder when $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ is divided by $(x - a)$ is $f(a)$.
- Let $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ and $f(a) = 0$ then $(x - a)$ is a factor of $f(x)$.
- If the degree of divisor is n then degree of remainder is $n - 1$.
- **Rational Fraction**: If $f(x)$ and $g(x)$ are two polynomials with $g(x) \neq 0$, then $\frac{f(x)}{g(x)}$ is called as a rational fraction.
- **Proper and Improper Fraction**: A rational fraction $\frac{f(x)}{g(x)}$ is called as a:
 1. Proper fraction if $\deg(f(x)) < \deg(g(x))$.
 2. Improper fraction if $\deg(f(x)) \geq \deg(g(x))$.
- **Division Algorithm**: Let $f(x)$ and $g(x)$ are two polynomials with $g(x) \neq 0$, then there exist two polynomials $q(x)$ and $r(x)$ such that $\frac{f(x)}{g(x)} = q(x) + \frac{r(x)}{g(x)}$ if $\deg(f(x)) > \deg(g(x)) > \deg(r(x))$.
- **Methods of resolving proper fraction $\frac{f(x)}{g(x)}$ into partial fractions.**
 1. $\frac{f(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$, where $A = \frac{f(a)}{a-b}$, $B = \frac{f(b)}{b-a}$.
 2. $\frac{f(x)}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$, where

$$A = \frac{f(a)}{(a-b)(a-c)}, B = \frac{f(b)}{(b-a)(b-c)}, C = \frac{f(c)}{(c-a)(c-b)}.$$
 3. $\frac{f(x)}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$, where

$$A = \frac{f(a)}{a-b} - \frac{f(a)}{(a-b)^2}, B = \frac{f(a)}{(a-b)^2}, C = \frac{f(b)}{(a-b)^2}.$$
 4. $\frac{f(x)}{(ax^2 + bx + c)(x-d)} = \frac{Ax+B}{ax^2 + bx + c} + \frac{C}{x-d}$ where $C = \frac{f(d)}{ad^2 + bd + c}$ and by equation of coefficient, we get A and B from $f(x) = (x-d)(Ax+B) + C(ax^2 + bx + c)$.
 5. $\frac{f(x)}{(ax^2 + bx + c)^2(x-d)} = \frac{Ax+B}{ax^2 + bx + c} + \frac{Cx+D}{(ax^2 + bx + c)^2} + \frac{E}{x-d}$ where

$$E = \frac{f(d)}{(ad^2 + bd + c)^2}$$
 and by equating the coefficients we get the value of A, B, C and D from $f(x) = (x-d)(ax^2 + bx + c)(Ax+B) + (Cx+D)(x-d) + E(ax^2 + bx + c)^2$.

$$6. \frac{px+q}{ax^2(x-a)} = -\frac{q}{ax^2} - \frac{pa+q}{a^2x} + \frac{pa+q}{a^2(x-a)}.$$

$$7. \frac{1}{x^3(x+a)} = \frac{1}{a^3x} - \frac{1}{a^2x^2} + \frac{1}{ax^3} - \frac{pa+q}{a^3(x-a)}.$$

$$8. \frac{1}{(x^2+b)(x-a)} = \frac{1}{a^2+b} \left(\frac{1}{x-a} - \frac{x+a}{x^2+b} \right).$$

$$9. \frac{1}{(x^2+a^2)(x^2+b^2)} = \frac{1}{b^2-a^2} \left(\frac{1}{x^2+a^2} - \frac{1}{x^2+b^2} \right) = \frac{1}{a^2-b^2} \left(\frac{1}{x^2+b^2} - \frac{1}{x^2+a^2} \right).$$